

Mathematics

A Cultural Heritage

Lecture 4

New Kinds of Math – Part 3:

Mathematics and the Geometry of Space

The Science of Space

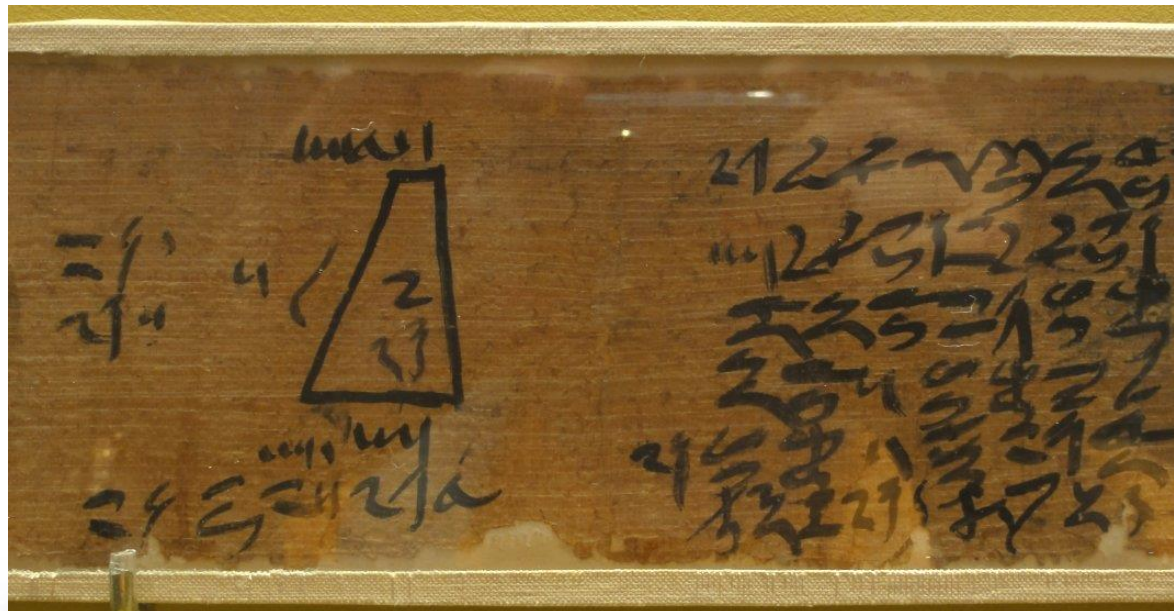
- In “*The Story of 1*”:
How ancient Egyptians measured space.



Scene from “The Story of 1”

Geometry in Ancient Civilizations

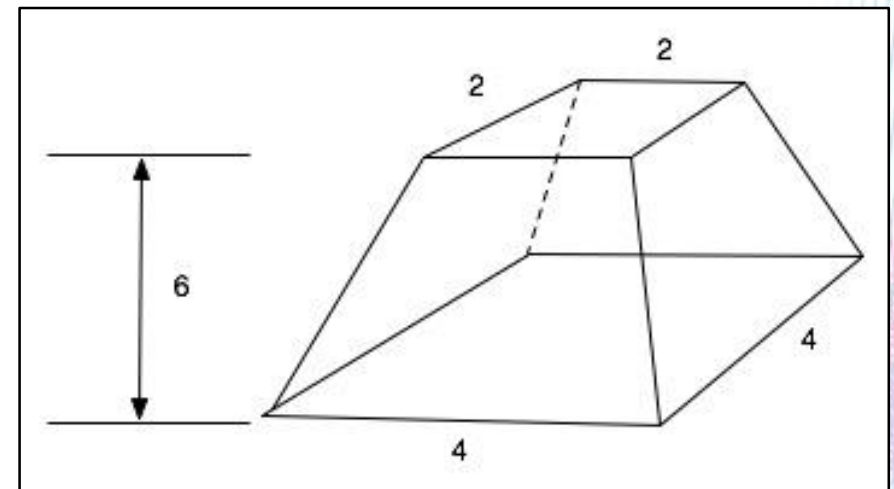
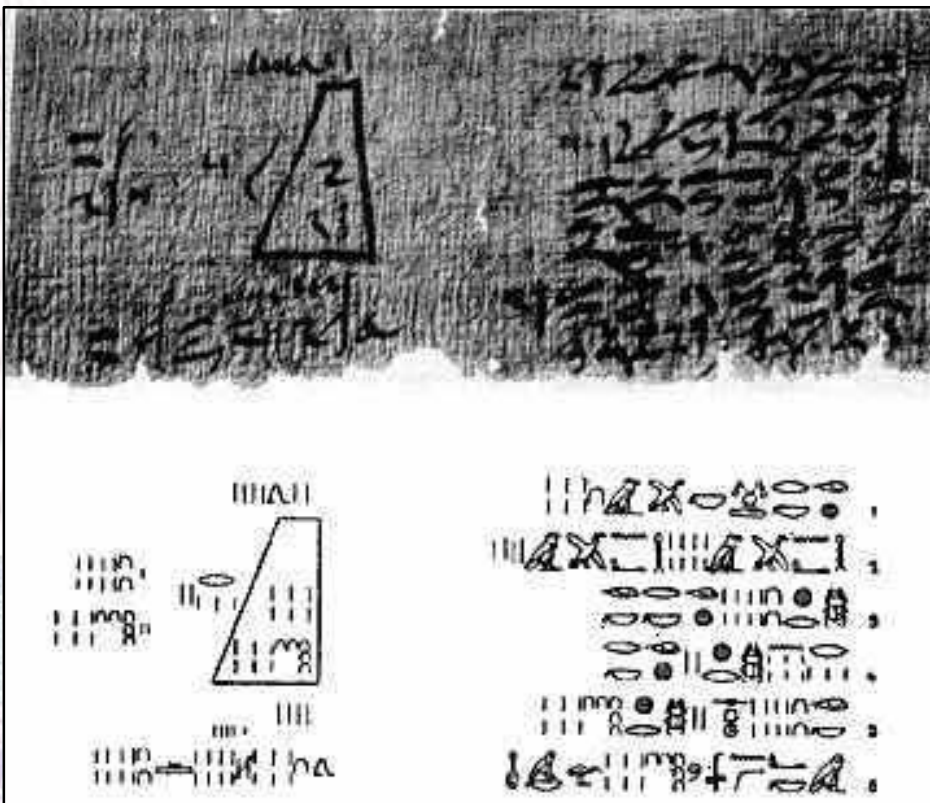
- The **Moscow Papyrus**
 - An **Egyptian papyrus** of more than 4000 years old
 - Measured 554 cm long, between 4 to 8 cm wide
 - Contains 25 math problems with solutions



Part of the Moscow Papyrus

The Moscow Papyrus

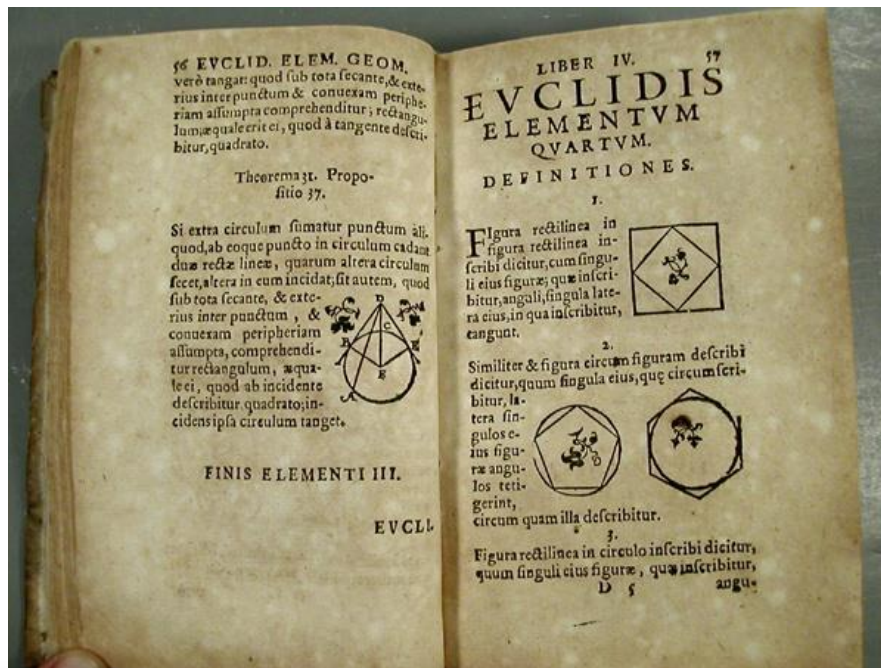
- Problem 14 in the Moscow Papyrus is finding the **volume** of a **truncated pyramid**.



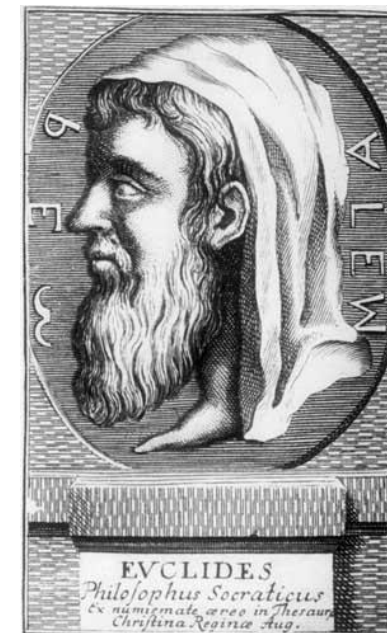
Modern drawing of the figure in Problem 14 of the Moscow Papyrus

Geometry and the Ancient Greeks

1. Word **geometry**: Greek **geo** (earth), **metrein** (measure)
2. Euclid's *Elements* was *the* Geometry textbook in the West for over 2000 years.



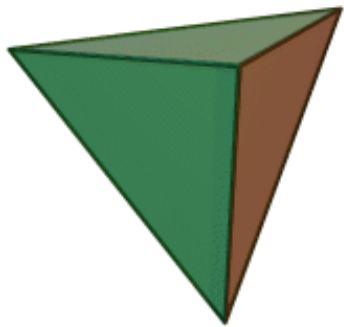
A later edition of *Elements*



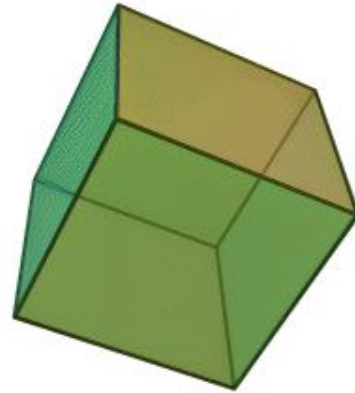
Euclid (323 – 283 BC)

Euclid's Elements

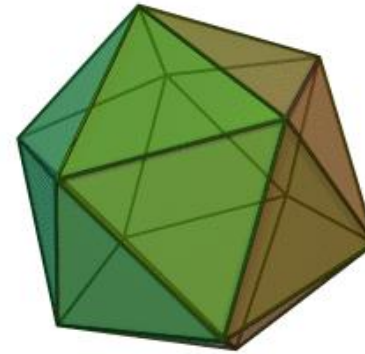
- **Platonic solids**
 - Each side regular polygon
 - Same number meeting at vertex



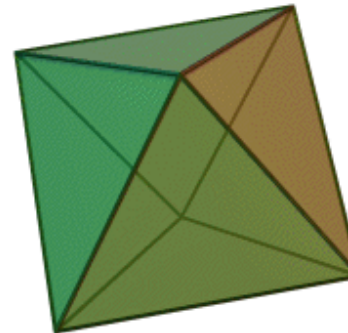
Tetrahedron



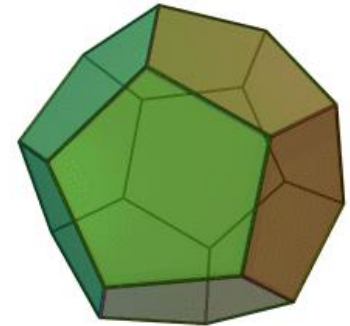
Cube



Icosahedron



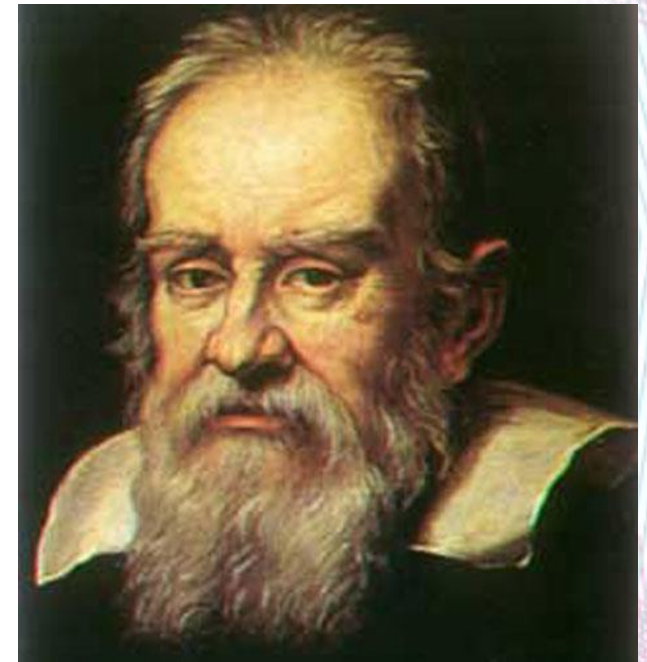
Octahedron



Dodecahedron

Geometry as Language to Describe the Universe

*“Philosophy [nature] is written in that great book which ever is before our eyes -- I mean the **universe** -- but we cannot understand it if we do not first learn the language and grasp the symbols in which it is written. The book is written in **mathematical language**, and the symbols are **triangles, circles** and other **geometrical figures**, without whose help it is impossible to comprehend a single word of it; without which one wanders in vain through a dark labyrinth.”*



Galileo Galilei (1564 - 1642)

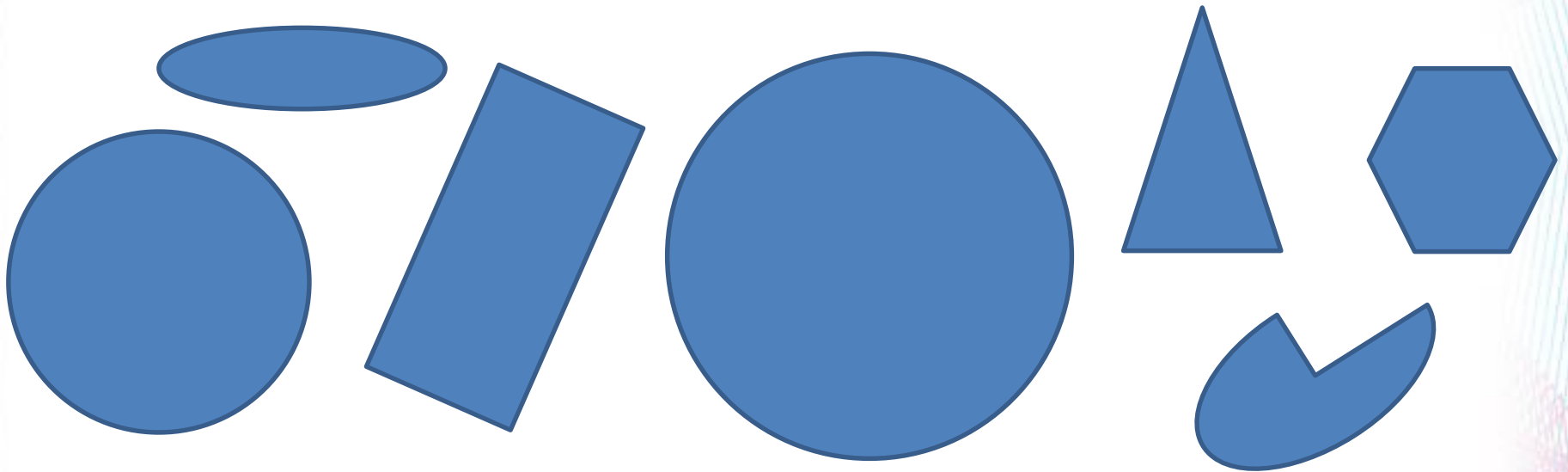
— Galileo Galilei

Topology

- Topology:
 - Study shapes
 - Allow (smooth) “stretching/bending” (deformations)
 - No gluing/cutting
- What is preserved? What is the essence?
 - Connected (one piece)
 - Number of “holes”?
 - ...

Usual Geometry vs. Topology

- Following figures clearly **different**:



- **Topologists**: all the same (**homeomorphic**).

Cups and Donuts

- Topologist: Coffee cups, donuts are the **same**!
- At least main properties are.

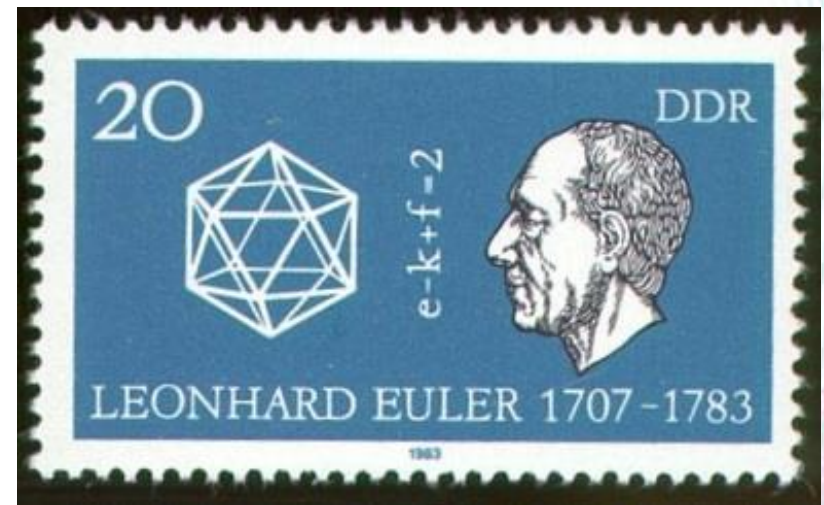
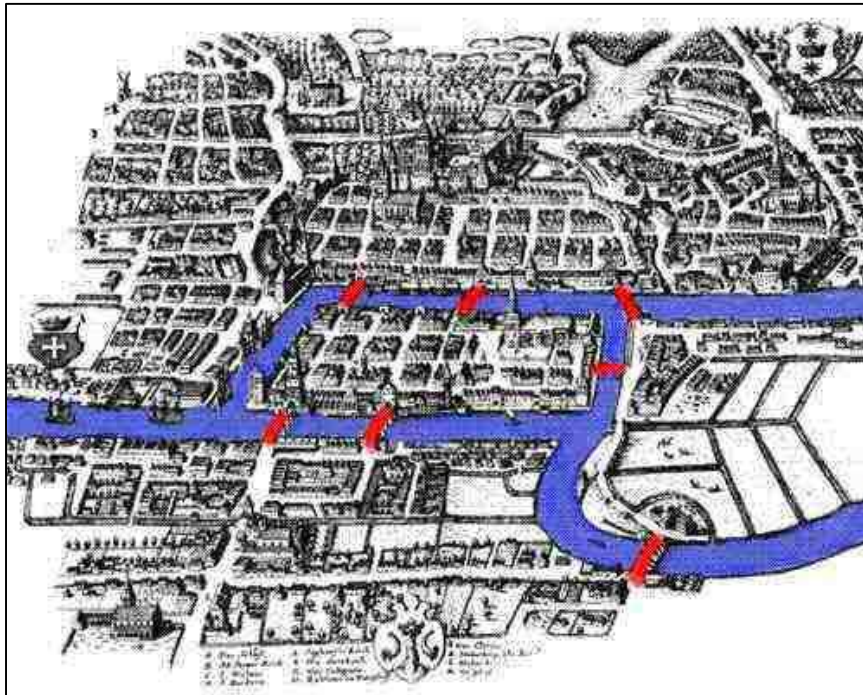


Coffee Cup, Donut, Glass, and Bun



The Seven Bridges of Königsberg

- **A puzzle:** Is it possible to walk through the city by crossing each bridge once and only once?

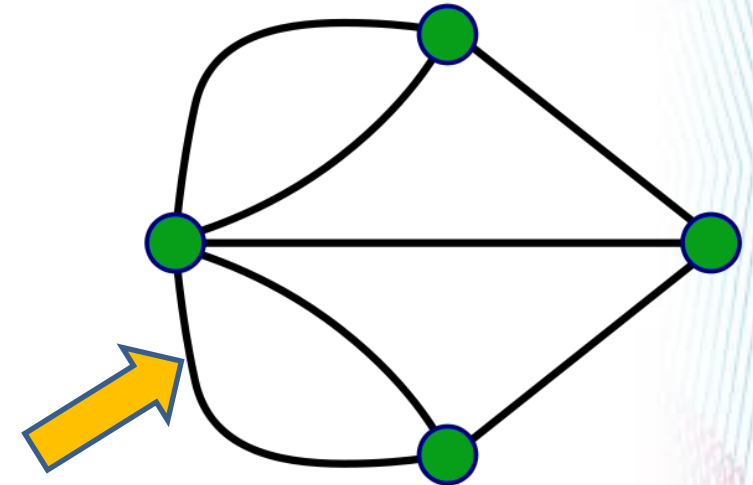
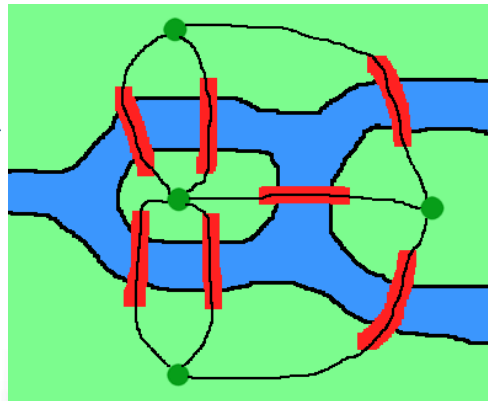
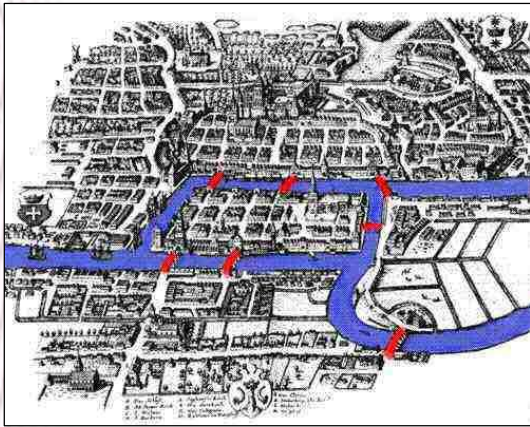


The formula in this stamp is another topological result by Euler

- **Leonhard Euler** (1735): impossible.

Key Step

- Key step: abstract view as **graph**:



Abstraction Process as a Way of Mathematics

- Reading assignment: “*Abstraction*”
 - Same process
 - Simpler with “visual cues” removed

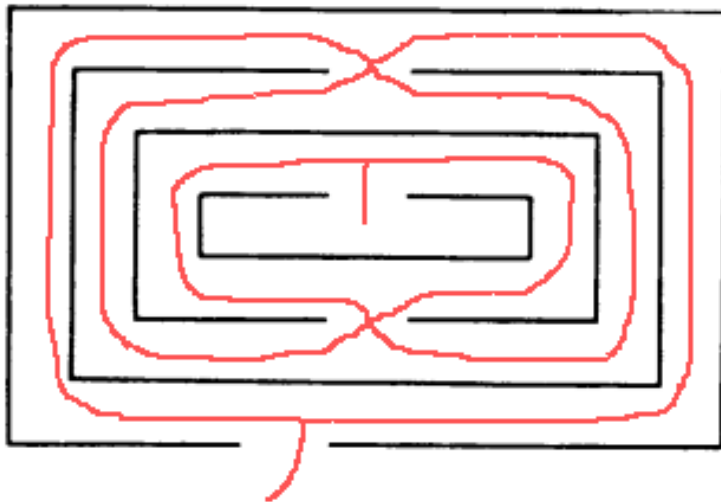


Figure 1.

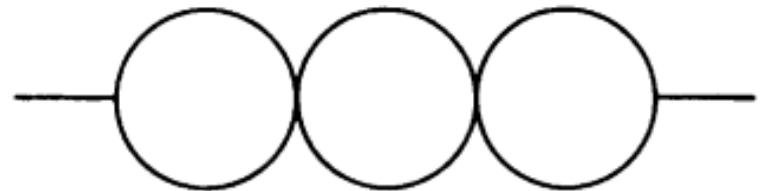


Figure 2.

Topology and Graph Theory

- Euler's solution gave two new branches:
 - Topology
 - Graph Theory.

